

Entropy principle.

$$dS \geq \frac{dQ}{T}$$

for isolated system $dQ = 0$

$$dS \geq 0$$

for reversible $dS = 0$

for irreversible $dS > 0$

→ Entropy may decrease locally at some region.

Applications of Entropy principle:

1. Transfer of Heat through a Finite temp. diff.
2. Mixing of Two fluids

$$m_1 c_1 (t_1 - t_f) = m_2 c_2 (t_f - t_2)$$

$$t_f = \frac{m_1 c_1 t_1 + m_2 c_2 t_2}{m_2 c_2 + m_1 c_1}$$

$$\Delta S_1 = \int_{T_1}^{T_f} \frac{dQ_{rev}}{T} = \int_{T_1}^{T_f} \frac{m_1 c_1 dT}{T} = m_1 c_1 \ln \left(\frac{T_f}{T_1} \right)$$

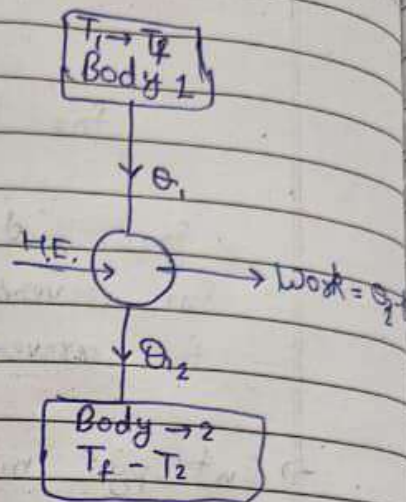
$$\Delta S_2 = \int_{T_2}^{T_f} \frac{m_2 c_2 dT}{T} = m_2 c_2 \ln \left(\frac{T_f}{T_2} \right)$$

1. Max. work Obtainable from two finite Bodies at temp. T_1 and T_2 :

$$T_f = \frac{T_1 + T_2}{2}$$

$$Q_1 = m c_p (T_1 - T_f) \quad T_1 > T_f > T_2$$

$$Q_2 = m c_p (T_f - T_2)$$



2. Max. work Obtainable from a finite Body and a TER.

$$\Delta S_{\text{Body}} = \int_T^{T_0} \frac{c_p dT}{T} - c_p \ln\left(\frac{T_0}{T}\right)$$

$$Q = m c_p (T - T_0)$$

$$\Delta S_1 = \int_T^{T_0} \frac{dQ}{T} = m c_p \ln\left(\frac{T_0}{T}\right)$$

$$\Delta S_2 = \frac{Q - W}{T_0}$$

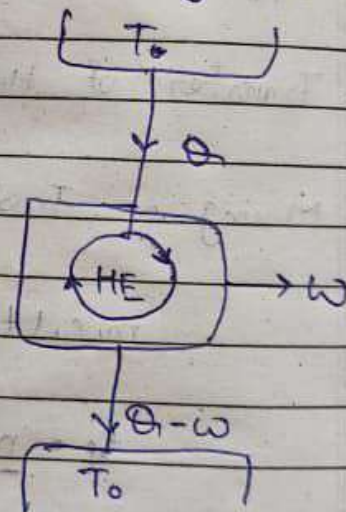
$$\Delta S_1 + \Delta S_2 \geq 0$$

$$\Delta S_{\text{HE}} = \int dS = 0$$

$$Q = c_p (T - T_0)$$

$$\Delta S_{\text{TER}} = \frac{Q - W}{T_0}$$

$$\Delta S_{\text{univ}} = c_p \ln \frac{T_0}{T} - \frac{Q - W}{T_0}$$



Finite Bodies

By the entropy principle

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$\rightarrow W_{\text{out}} = Q_2 - Q_1$

$$C_p \ln \frac{T_0}{T} > \frac{W - Q}{T_0}$$

$$\frac{W - Q}{T_0} \leq C_p \ln (T_0/T)$$

a TER.

$$W \leq 0 + T_0 C_p \ln (T_0/T)$$

$$W_{\text{max}} = Q + T_0 C_p \ln (T_0/T)$$

Combined consequences of first and second law

$$d\Theta_R = T ds$$

$$d\Theta = dU + p dv \quad \text{Closed, non flow}$$

$$T ds = dU + p dv$$

Enthalpy

$$H = U + pV$$

$$dH = dU + p dv + v dp$$

$$dH = T ds + v dp$$